

# 2018-04-11 TRAFFIC ANALYSIS EXERCISE: ANSWERS

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## DYNACCOUNTIC

[www.malware-traffic-analysis.net/2018/04/11/2018-04-11-traffic-analysis-exercise.pcap.zip](http://www.malware-traffic-analysis.net/2018/04/11/2018-04-11-traffic-analysis-exercise.pcap.zip)

The zip file is password-protected with the standard password. See the "about" page of my site if you don't know it.

The associated network segment:

- LAN segment: 10.10.10.0/24 (10.10.10.0 through 10.10.10.255)
- Broadcast address: 10.10.10.255
- Domain controller: 10.10.10.3 (DYNACCOUNTIC-DC)
- Domain: dynaccountic.com

### YOUR TASK:

Write an incident report that includes:

- Date/Time of the infection
- Who was infected (IP address, host name, MAC address, and user account name)
- What malware is involved
- The likely source of this infection
- Indicators associated with this infection (IP addresses, domains, URLs, and file hashes, if any)

## ANSWERS

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### SUMMARY:

On Tuesday 2018-04-10 at 20:14 UTC, a Windows computer used by winford.batiste was infected with Trickbot malware. This infection likely originated from an malicious spam as described in an article on My Online Security (see below for link). The infected computer was sent to our help desk to be wiped and re-imaged. Winford Batiste has canged all of his login credentials for any accounts he used on that computer.

### DETAILS:

- IP address: 10.10.10.209
- MAC addres: 00:30:67:Fa:2d:63
- Host name: Batiste-PC
- User account name: winford.batiste
- My Online Security's blog post on the associated malspam is available at:  
<https://myonlinesecurity.co.uk/another-fake-sage-outdated-invoice-scam-delivering-trickbot-via-microsoft-equation-editor-exploits-cve-2017-11882-and-cve-2017-8570/>

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### INDICATORS:

- 95.110.193.132 port 80 - **caveaudelteatro.it** - GET /ser0410.bin - *returned Trickbot binary*
- port 80 - **myexternalip.com** - GET /raw - *IP address check by the infected host*
- 82.214.141.134 port 449 - *SSL/TLS traffic generated by Trickbot*
- 86.61.160.50 port 447 - *SSL/TLS traffic generated by Trickbot*

SHA256 hash: [c2c1e2c22f67dda6553cbcc173694b68677b77319243684925e8dc3f78b3dbf8](#)

- File size: 364,544 bytes
- File description: Trickbot binary downloaded from caveaudelteatro.it

### DETAILS

As always, I recommend you set up Wireshark according to the tutorials I've documented at:

- <http://malware-traffic-analysis.net/tutorials/index.html>

I submitted the pcap to both VirusTotal ([link](#)) and Packettotal ([link](#)). Packettotal didn't have anything immediately obvious, and you have to wait a while on VirusTotal for the alerts to show up (if they ever do). In this case, I've already run the exercise pcap through my Suricata/ETPRO setup on Security Onion, where it gave me the following alerts:

| RealTime Events Escalated Events |     |               |                |       |               |       |    |                                                             |
|----------------------------------|-----|---------------|----------------|-------|---------------|-------|----|-------------------------------------------------------------|
| ST                               | CNT | Date/Time     | Src IP         | SPort | Dst IP        | DPort | Pr | Event Message                                               |
| RT                               | 2   | 2018-04-10... | 95.110.193.132 | 80    | 10.10.10.209  | 49199 | 6  | ET POLICY Binary Download Smaller than 1 MB Likely Hostile  |
| RT                               | 2   | 2018-04-10... | 95.110.193.132 | 80    | 10.10.10.209  | 49199 | 6  | ET POLICY PE EXE or DLL Windows file download HTTP          |
| RT                               | 2   | 2018-04-10... | 95.110.193.132 | 80    | 10.10.10.209  | 49199 | 6  | ET INFO Executable Retrieved With Minimal HTTP Headers -... |
| RT                               | 1   | 2018-04-10... | 10.10.10.209   | 49200 | 78.47.139.102 | 80    | 6  | ET POLICY Possible IP Check myexternalip.com                |
| RT                               | 3   | 2018-04-10... | 82.214.141.134 | 449   | 10.10.10.209  | 49206 | 6  | ET POLICY OpenSSL Demo CA - Internet Widgits Pty (O)        |
| RT                               | 4   | 2018-04-10... | 86.61.160.50   | 447   | 10.10.10.209  | 49207 | 6  | ETPRO TROJAN Trickbot SSL Certificate Detected              |

Shown above: Some alerts on the infection traffic from my Suricata/ETPRO setup on Security Onion.

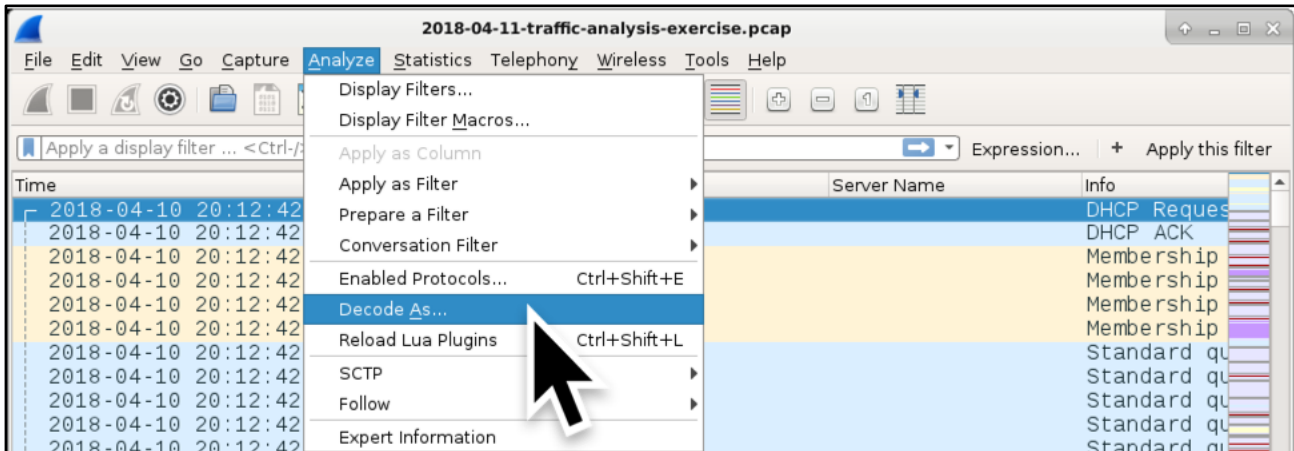
The last alert in the above list is for Trickbot. This matches what I've seen for Trickbot before.

| http.request or ssl.handshake.type == 1 |                |      |                    |                      |                    |  |
|-----------------------------------------|----------------|------|--------------------|----------------------|--------------------|--|
| Time                                    | Dst            | port | Host               | Server Name          | Info               |  |
| 2018-04-10 20:12:53                     | 23.43.62.200   | 80   | www.msftncsl.com   |                      | GET /ncsl.txt HTTP |  |
| 2018-04-10 20:14:26                     | 95.110.193.132 | 80   | caveaudelteatro.it |                      | GET /ser0410.bin H |  |
| 2018-04-10 20:14:40                     | 78.47.139.102  | 80   | myexternalip.com   |                      | GET /raw HTTP/1.1  |  |
| 2018-04-10 20:46:32                     | 82.214.141.134 | 449  |                    | clients2.google.c... | Client Hello       |  |
| 2018-04-10 20:46:34                     | 86.61.160.50   | 447  |                    | update.googleapis... | Client Hello       |  |
| 2018-04-10 21:50:39                     | 86.61.160.50   | 447  |                    |                      | Client Hello       |  |
| 2018-04-10 22:21:11                     | 86.61.160.50   | 447  |                    |                      | Client Hello       |  |
| 2018-04-10 22:51:14                     | 82.214.141.134 | 449  |                    |                      | Client Hello       |  |
| 2018-04-10 22:54:35                     | 82.214.141.134 | 449  |                    |                      | Client Hello       |  |
| 2018-04-10 22:56:41                     | 86.61.160.50   | 447  |                    |                      | Client Hello       |  |
| 2018-04-10 23:27:02                     | 86.61.160.50   | 447  |                    |                      | Client Hello       |  |

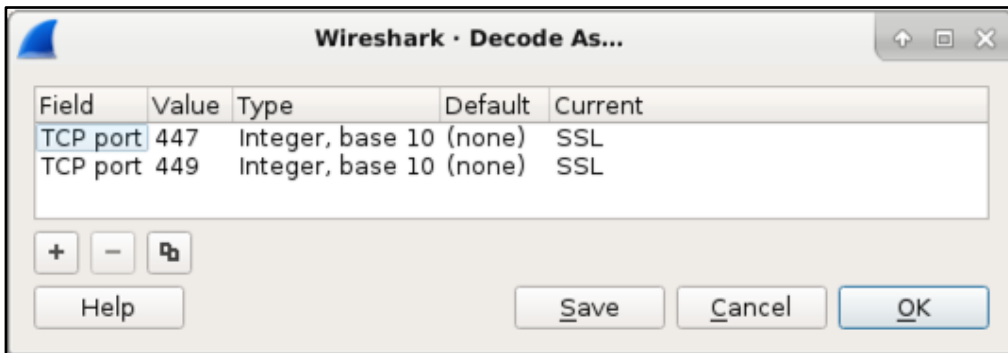
Shown above: Filtering on the traffic in Wireshark.

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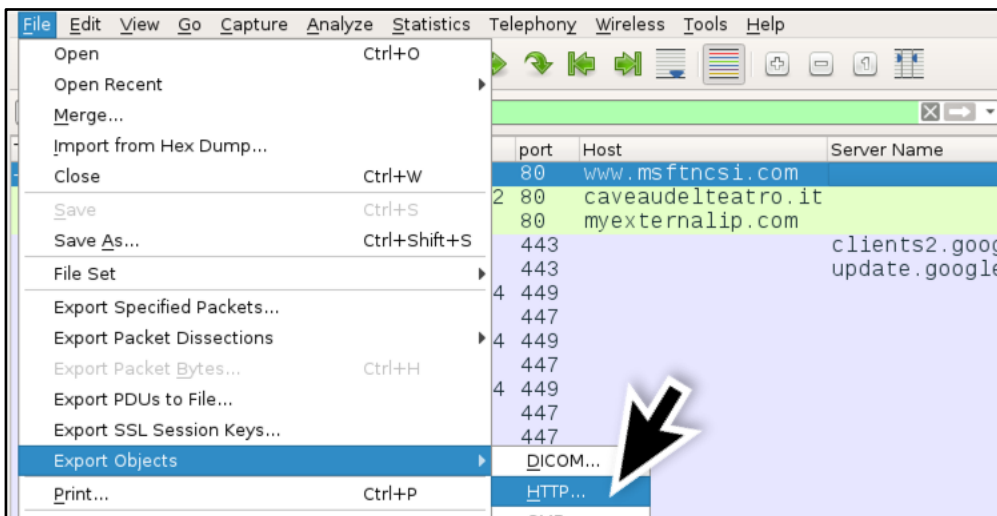
Of note, in the above image, I've set the decoding for TCP ports 447 and TCP ports 449 for SSL in Wireshark. It's easy enough to do. From the Wireshark menu, use **Analyze → Decode As...**



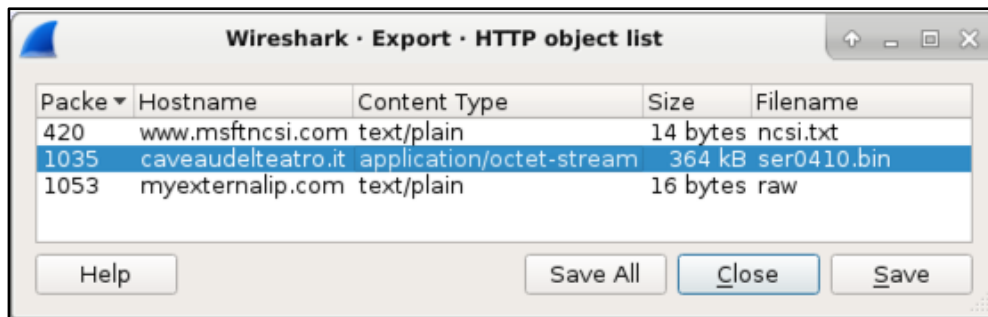
Then add the appropriate TCP ports to decode as SSL. The **Decode As...** window should look like this when you're done:



Let's try another trick, and see if there's anything interesting we can export from the pcap. From the Wireshark menu, use **File → Export Object → HTTP...**



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It looks like an application/octet-stream was returned from **caveaudelteatro.it** with an HTTP request ending in **ser0410.bin**. Let's select that file and save it as **ser0410.bin**.

I'm generally working in a Linux environment when I review these pcaps, so I can just do a file command to see what type of file it is, then I can get the SHA256 hash and check it in VirusTotal.

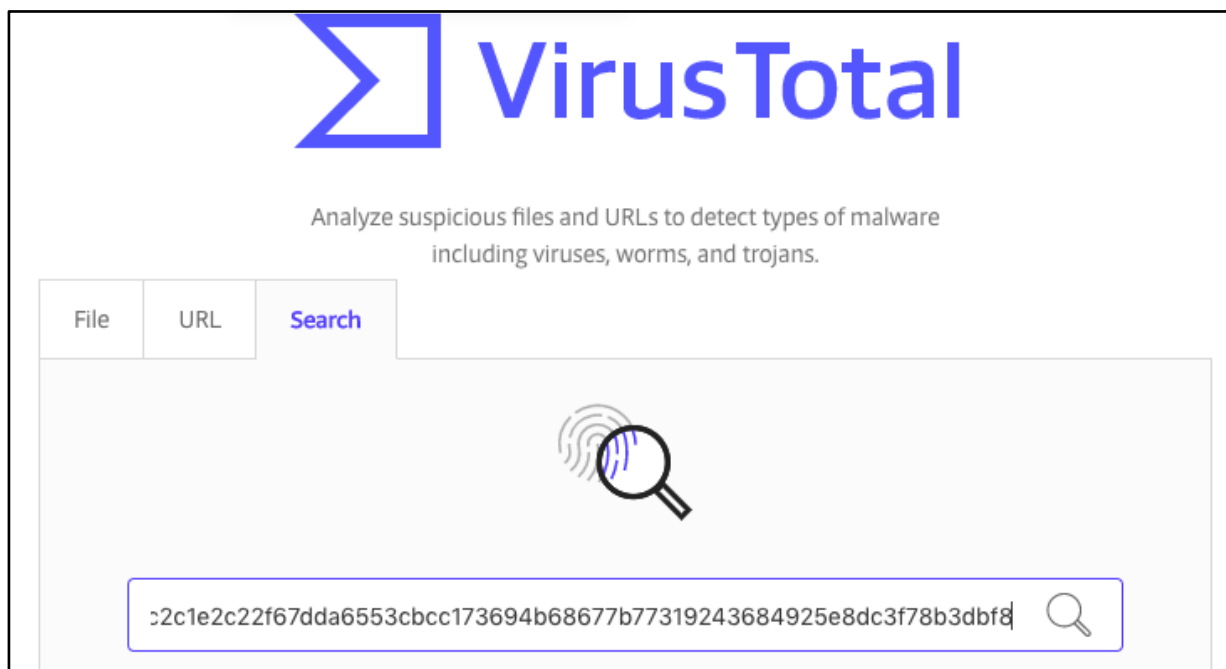
```
$ file ser0410.bin
```

```
ser0410.bin: PE32 executable (GUI) Intel 80386, for MS Windows
```

```
$ shasum -a 256 ser0410.bin
```

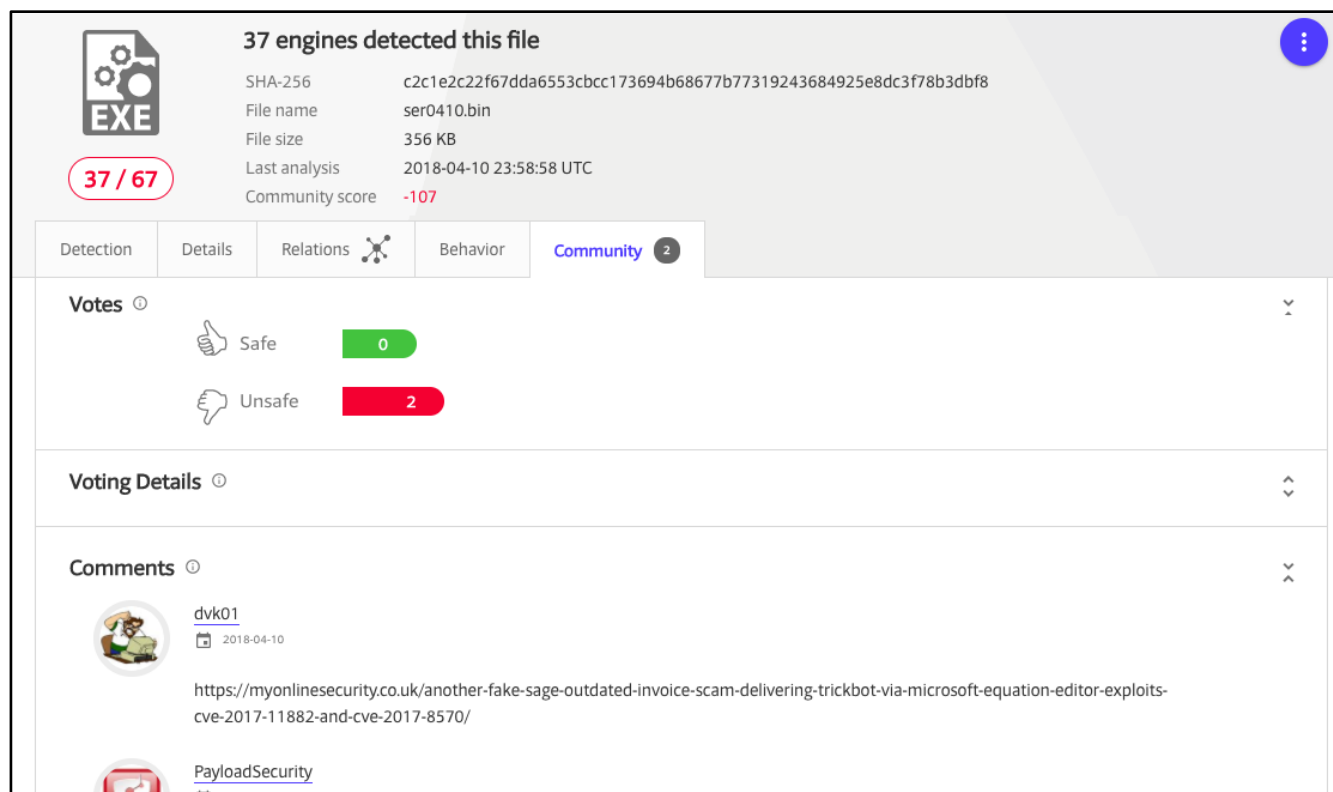
```
c2c1e2c22f67dda6553cbcc173694b68677b77319243684925e8dc3f78b3dbf8  
ser0410.bin
```

Searching the SHA256 hash in virustotal.com should reveal some comments stating this is Trickbot.



Shown above: Searching the SHA256 hash of **ser0410.bin** on virustotal.com.

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The screenshot shows the VirusTotal interface for a file named **ser0410.bin**. The file is identified as an **EXE** (Executable) with a SHA-256 hash of **c2c1e2c22f67dda6553bcc173694b68677b77319243684925e8dc3f78b3dbf8**. It is 356 KB in size and was last analyzed on 2018-04-10 at 23:58:58 UTC. The community score is **-107**. The interface shows 37 engines detected this file. The **Community** tab is active, showing 2 votes: 0 Safe and 2 Unsafe. A comment from user **dvk01** on 2018-04-10 includes a link to a My Online Security article about Trickbot malware. The comment text is: <https://myonlinesecurity.co.uk/another-fake-sage-outdated-invoice-scam-delivering-trickbot-via-microsoft-equation-editor-exploits-cve-2017-11882-and-cve-2017-8570/>. The comment is from **PayloadSecurity**.

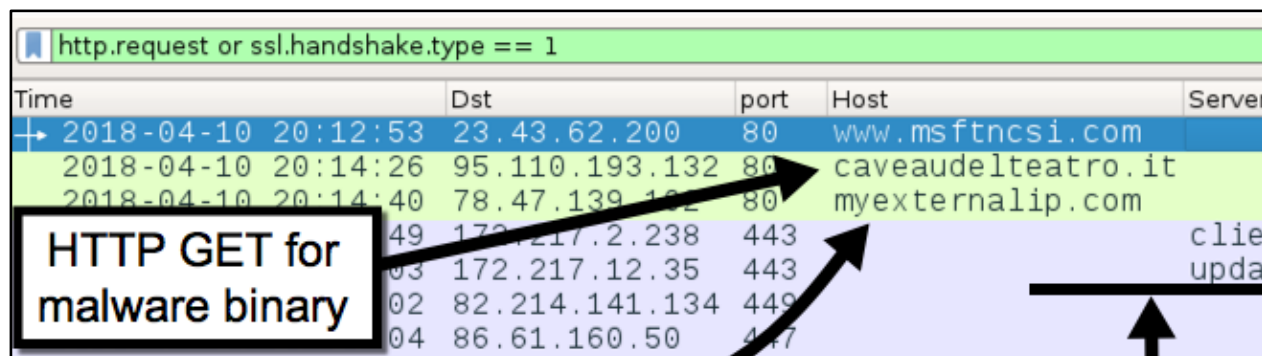
Shown above: Comments for this sample include a link to My Online Security for Trickbot malspam.

So we can say this is probably Trickbot infection based on the alerts and comments in VirusTotal. I definitely recognize this traffic, since I've seen Trickbot so often. I've documented relatively recent example of Trickbot at <https://www.malware-traffic-analysis.net/2018/03/22/index.html>.

Feel free to review that blog post and see the similarities in the traffic. If you see it often enough, it's easily recognizable.

Now that we've established the "what" for this exercise, let's determine the "when" and the "who." After all, it doesn't matter what happened, if we don't know when it happened or who it happened to.

When is easy enough to determine, based on the HTTP request for that Trickbot binary from **caveaudeleteatro.it**. It happened on 2018-04-10 at 20:14 UTC.



The screenshot shows a Wireshark packet capture of an HTTP GET request. The packet list on the left shows three packets: a GET request to **www.msftncsi.com** at 20:12:53, a GET request to **caveaudeleteatro.it** at 20:14:26, and a GET request to **myexternalip.com** at 20:14:40. The packet details pane on the right shows the **HTTP GET for malware binary** request. The packet bytes pane on the right shows the **clie** and **upda** fields. An arrow points from the **HTTP GET for malware binary** text box to the **caveaudeleteatro.it** entry in the packet list.

| Time                | Dst            | port | Host                | Server |
|---------------------|----------------|------|---------------------|--------|
| 2018-04-10 20:12:53 | 23.43.62.200   | 80   | www.msftncsi.com    |        |
| 2018-04-10 20:14:26 | 95.110.193.132 | 80   | caveaudeleteatro.it |        |
| 2018-04-10 20:14:40 | 78.47.139.182  | 80   | myexternalip.com    |        |
| 49                  | 172.217.2.238  | 443  |                     | clie   |
| 03                  | 172.217.12.35  | 443  |                     | upda   |
| 02                  | 82.214.141.134 | 449  |                     |        |
| 04                  | 86.61.160.50   | 447  |                     |        |

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This pcap was traffic pulled from a single IP address, which is **10.10.10.209** as you can quickly find. Just check what MAC address is using 10.10.10.209, then filter on NBNS traffic to figure out the host name. Sometimes, you can also look at the DHCP traffic (**udp.port eq 67** or **udp.port eq 68**) to get a host name.

The image shows a Wireshark packet capture of NetBIOS Name Service (NBNS) traffic. The packet list at the top shows five packets from 10.10.10.209 to 10.10.10.1 on port 137. The details pane for the selected packet (Frame 13) shows the structure of a NetBIOS Name Service registration request. The request includes a transaction ID, flags, and a query for the host name BATISTE-PC. The IP address 10.10.10.209 is highlighted in the details pane.

| Time                | Src          | port | Dst        | port | Info                           |
|---------------------|--------------|------|------------|------|--------------------------------|
| 2018-04-10 20:12:42 | 10.10.10.209 | 137  | 10.10.10.1 | 137  | Registration NB BATISTE-PC<00> |
| 2018-04-10 20:12:42 | 10.10.10.209 | 137  | 10.10.10.1 | 137  | Registration NB DYNACCOUNTIC<  |
| 2018-04-10 20:12:42 | 10.10.10.209 | 137  | 10.10.10.1 | 137  | Registration NB BATISTE-PC<20  |
| 2018-04-10 20:12:43 | 10.10.10.209 | 137  | 10.10.10.1 | 137  | Name query NB ISATAP<00>       |
| 2018-04-10 20:12:44 | 10.10.10.209 | 137  | 10.10.10.1 | 137  | Registration NB BATISTE-PC<20  |

Frame 13: 110 bytes on wire (880 bits), 110 bytes captured (880 bits)  
Ethernet II, Src: BiostarM\_f1:2d:63 (00:30:67:f1:2d:63), Dst: Cisco\_fe:c2:8e (00:09:7...)  
Internet Protocol Version 4, Src: 10.10.10.209, Dst: 10.10.10.1  
User Datagram Protocol, Src Port: 137, Dst Port: 137  
NetBIOS Name Service  
Transaction ID: 0x9ab1  
Flags: 0x2900, Opcode: Registration, Recursion desired  
Questions: 1  
Answer RRs: 0  
Authority RRs: 0  
Additional RRs: 1  
Queries  
BATISTE-PC<00>: type NB, class IN  
Additional records  
BATISTE-PC<00>: type NB, class IN  
Name: BATISTE-PC<00> (Workstation/Redirector)  
Type: NB (32)  
Class: IN (1)  
Time to live: 3 days, 11 hours, 20 minutes  
Data length: 6  
Name flags: 0x6000, ONT: Unknown (H-node, unique)  
Addr: 10.10.10.209

Shown above: Correlating the IP address, MAC address and host name in NBNS traffic.

Since this is a Windows environment, Kerberos traffic might have a user account name. In this case, it does. Use the following filter in Wireshark:

***ip.addr eq 10.10.10.209 and kerberos.CNameString and !(kerberos.CNameString contains \$)***

As always, this is for the most recent versions (anything 2.x). I strongly encourage people to use the most recent version of Wireshark if possible. For older versions of Wireshark (1.12 and lower), use the following filter:

***ip.addr eq 10.10.10.209 and kerberos.cname\_element and kerberos.KerberosString and !(kerberos.KerberosString contains \$)***

This quickly shows us 10.10.10.209 is **winford.batiste** on **BATISTE-PC**.

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Filter: `ip.addr eq 10.10.10.209 and kerberos.CNameString and !(kerberos.CNameString contains $)`

| Time                | Src          | port   | Dst           | port   | Info    |
|---------------------|--------------|--------|---------------|--------|---------|
| 2018-04-10 20:13:30 | 10.10.10.209 | 491... | 10.10.10.3    | 88     | AS-REQ  |
| 2018-04-10 20:13:30 | 10.10.10.209 | 491... | 10.10.10.3    | 88     | AS-REQ  |
| 2018-04-10 20:13:30 | 10.10.10.3   | 88     | 10.10.10.2... | 491... | AS-REP  |
| 2018-04-10 20:13:30 | 10.10.10.3   | 88     | 10.10.10.2... | 491... | TGS-REP |
| 2018-04-10 20:13:30 | 10.10.10.3   | 88     | 10.10.10.2... | 491... | TGS-REP |

Frame 452: 293 bytes on wire (2344 bits), 293 bytes captured (2344 bits)

- Ethernet II, Src: BiostarM\_f1:2d:63 (00:30:67:f1:2d:63), Dst: Dell\_ae:a5:4d (90:b1:)
- Internet Protocol Version 4, Src: 10.10.10.209, Dst: 10.10.10.3
- Transmission Control Protocol, Src Port: 49183, Dst Port: 88, Seq: 1, Ack: 1, Len:
- Kerberos
  - Record Mark: 235 bytes
  - as-req
    - pvno: 5
    - msg-type: krb-as-req (10)
    - padata: 1 item
    - req-body
      - Padding: 0
      - kdc-options: 40810010 (forwardable, renewable, canonicalize, renewable-ok)
      - cname
        - name-type: KRB5-NT-PRINCIPAL (1)
        - cname-string: 1 item
          - CNameString: winford.batiste
      - realm: DYNACCOUNTING
      - sname
      - time: 2027-09-12 02:48:05 (UTC)

Windows user account name

Now we know the when, who, and what. So the incident report should be relatively simple. This month's exercise is certainly more straight-forward than last month's exercise.

As always, there are many methods of investigating this pcap, and they are too numerous to include here.